North Fork St. Lucie Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

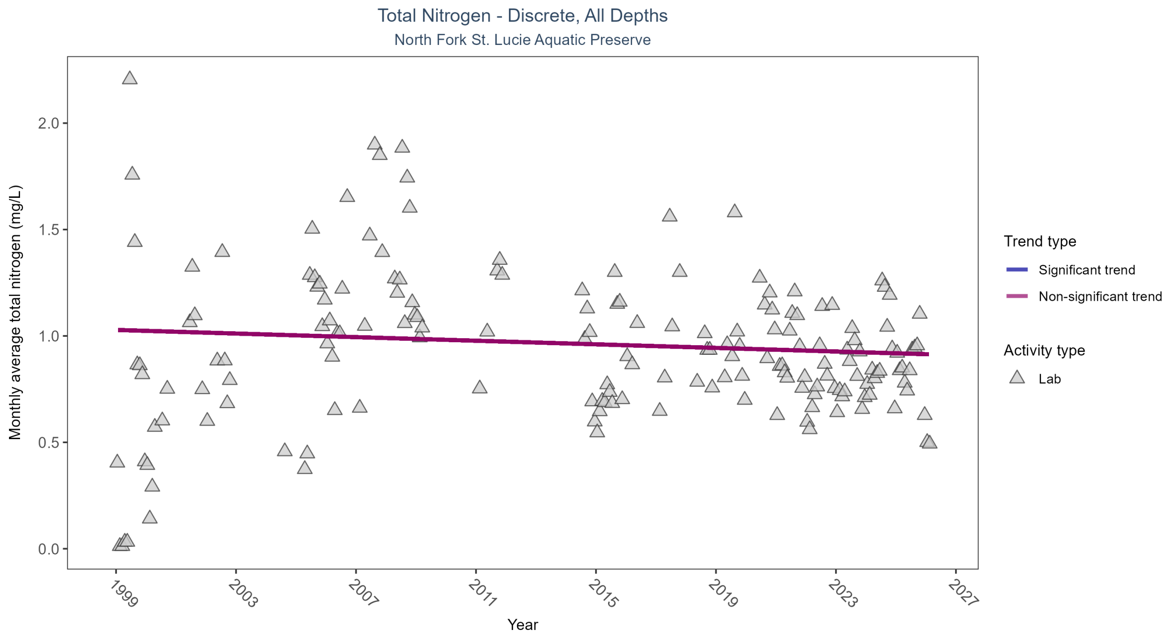

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1999 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	636	24	1999 - 2026	0.964	-0.08099	1.02814	-0.00422	0.1244

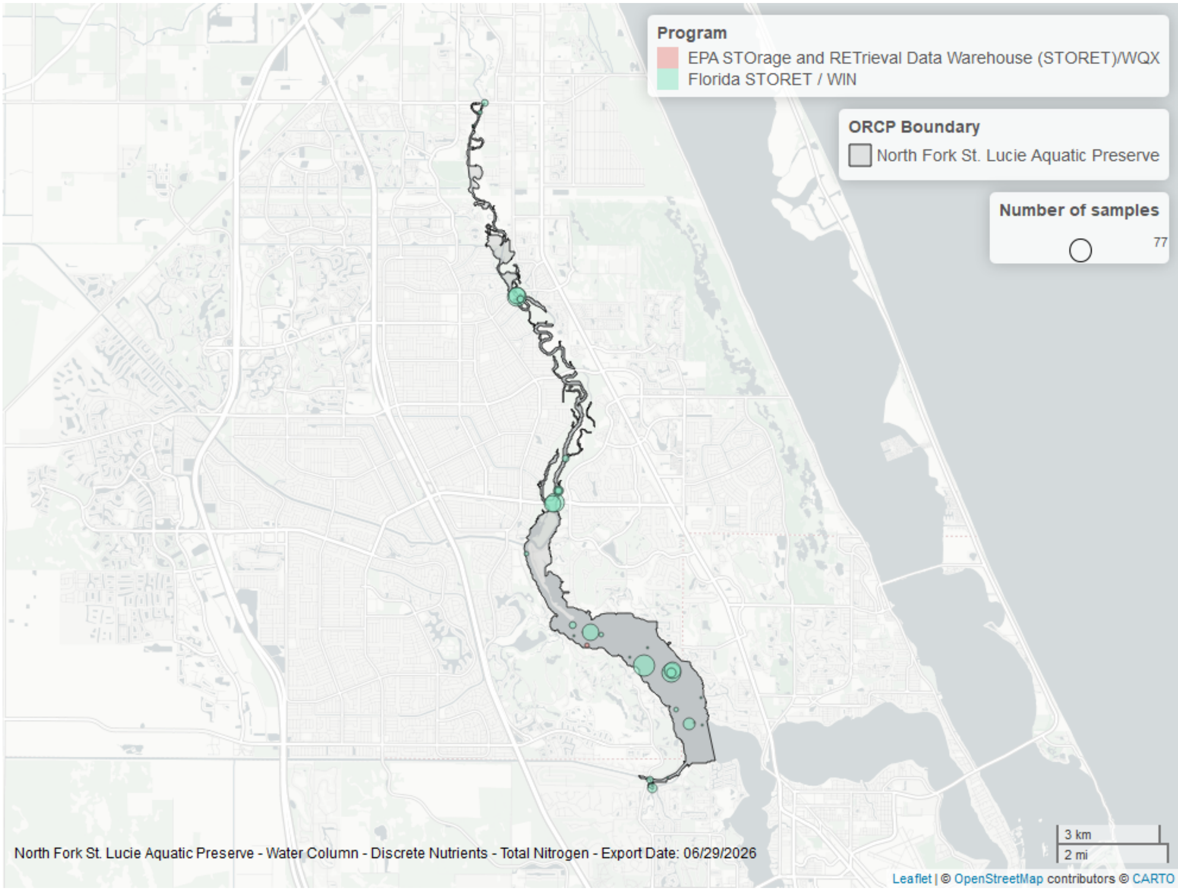

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

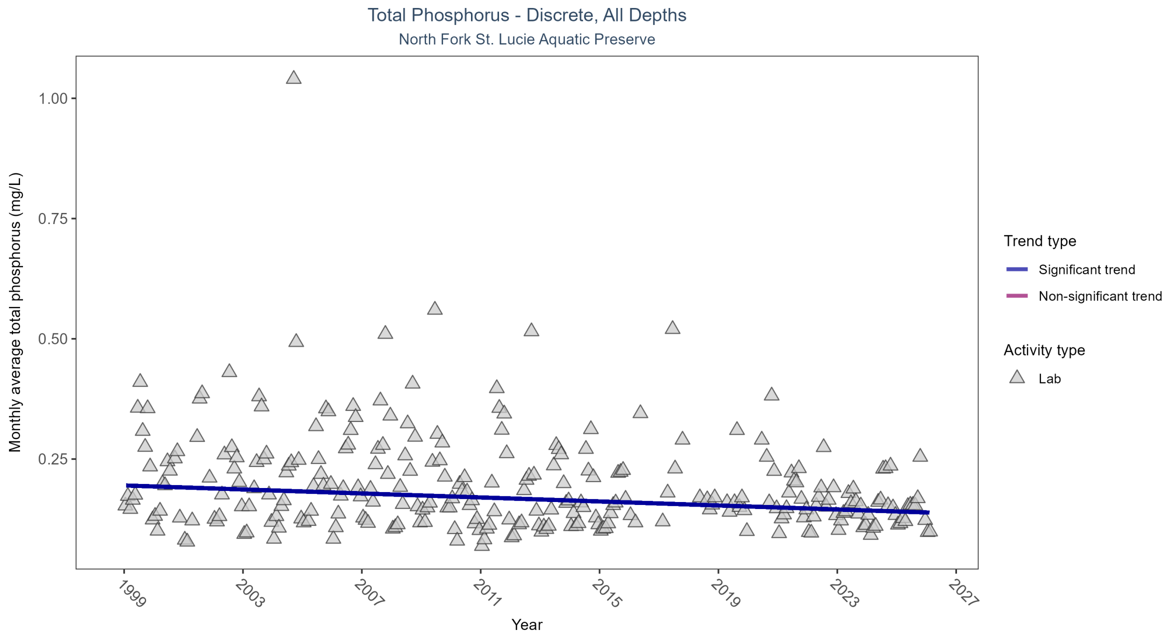

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	1075	28	1999 - 2026	0.175	-0.27455	0.19513	-0.00208	0

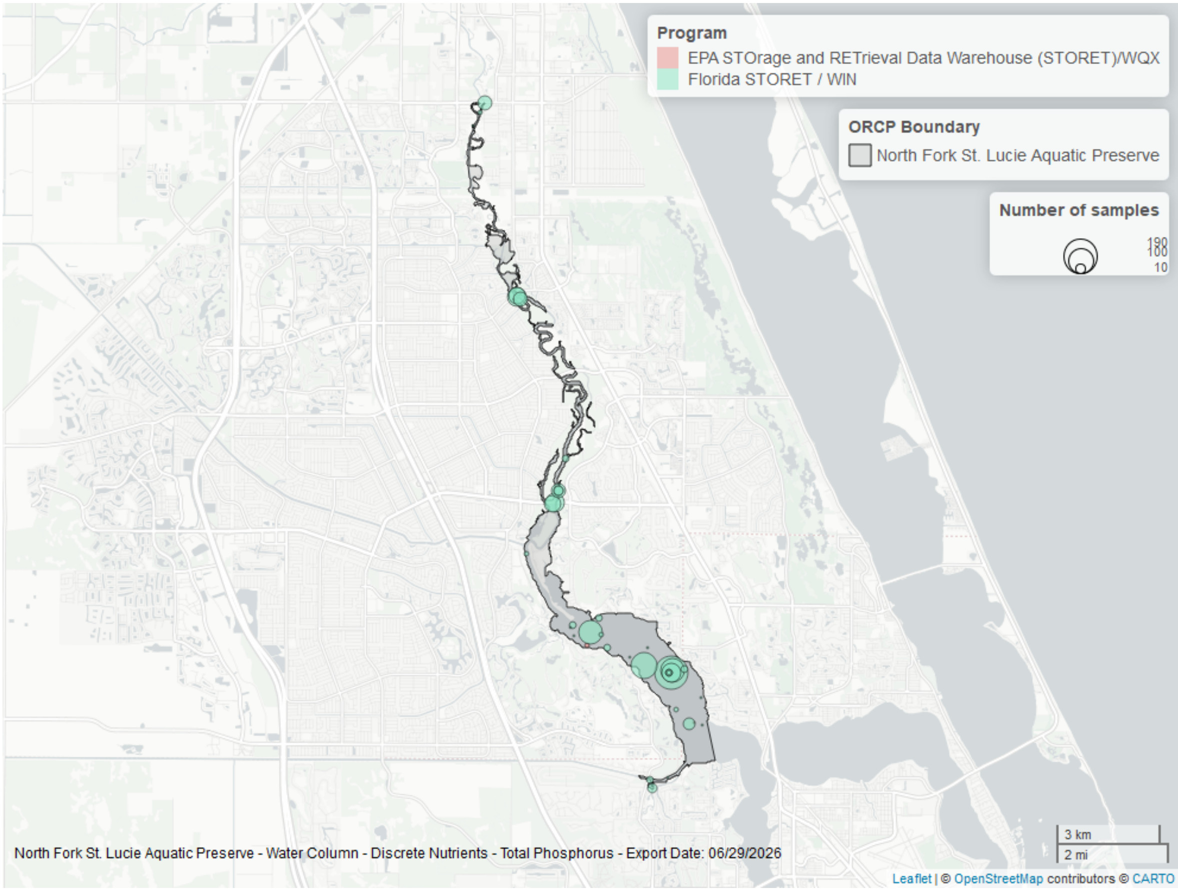

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Discrete

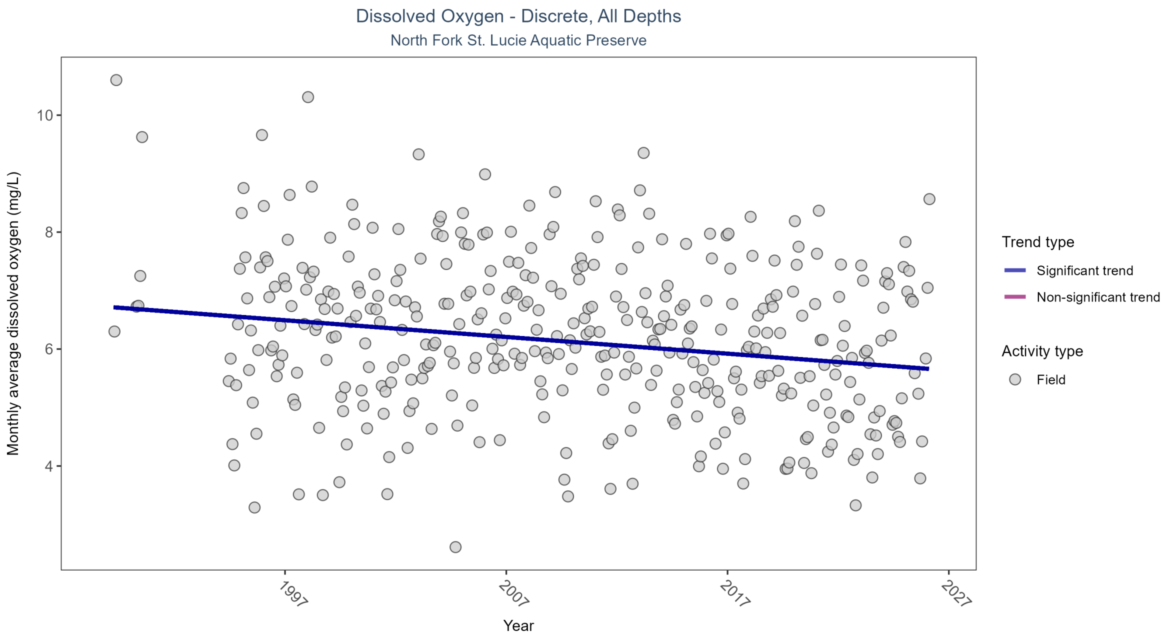

Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 3: Monthly average dissolved oxygen decreased by 0.03 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	7582	35	1989 - 2026	6.19	-0.21723	6.72097	-0.02857	0

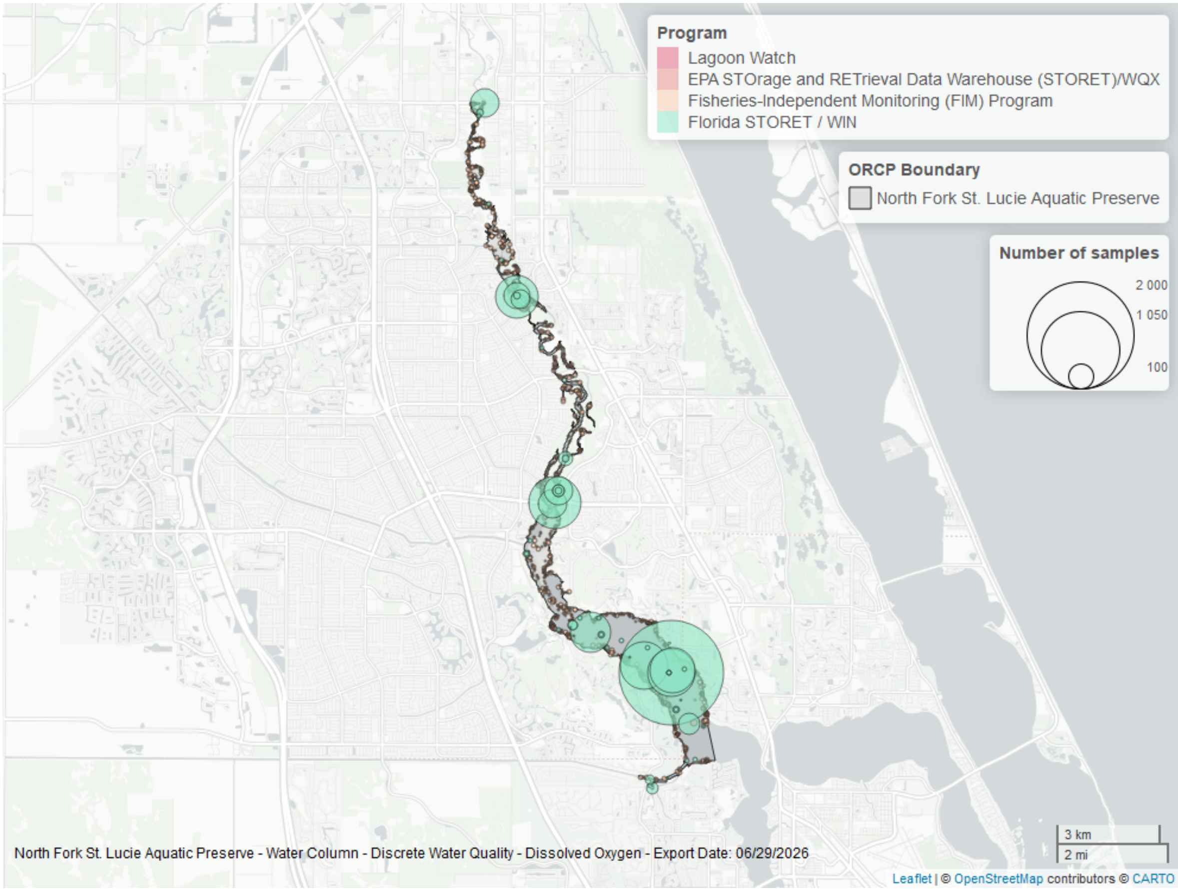

Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

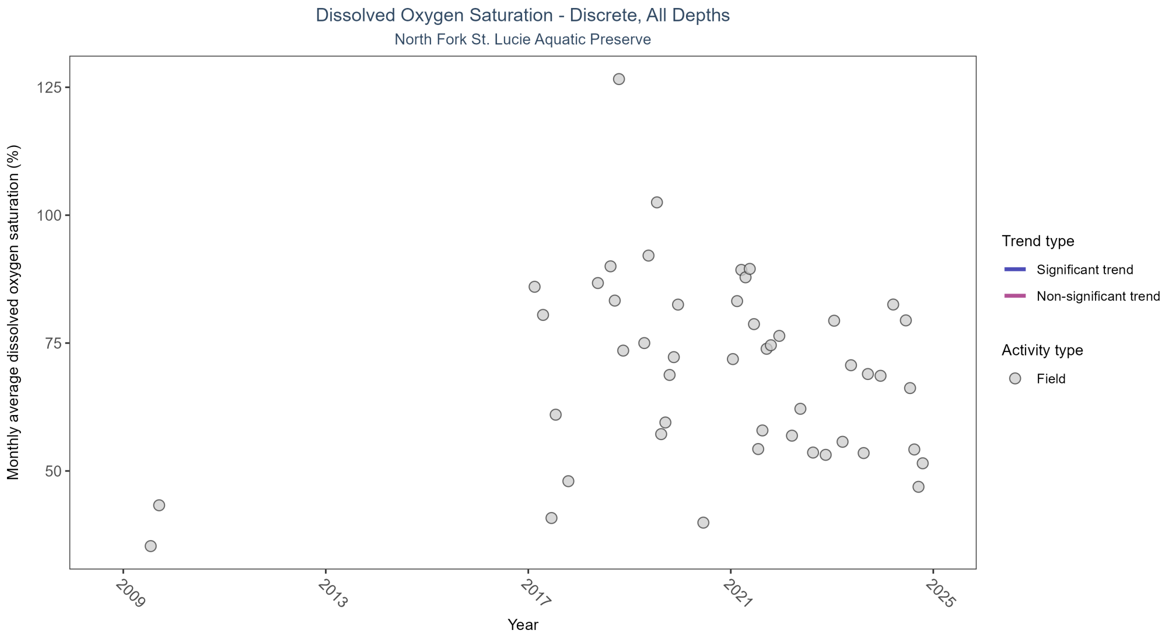

Figure 7: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 4: There was insufficient data to fit a model for dissolved oxygen saturation.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data	149	9	2009 - 2024	70.7	-	-	-	-

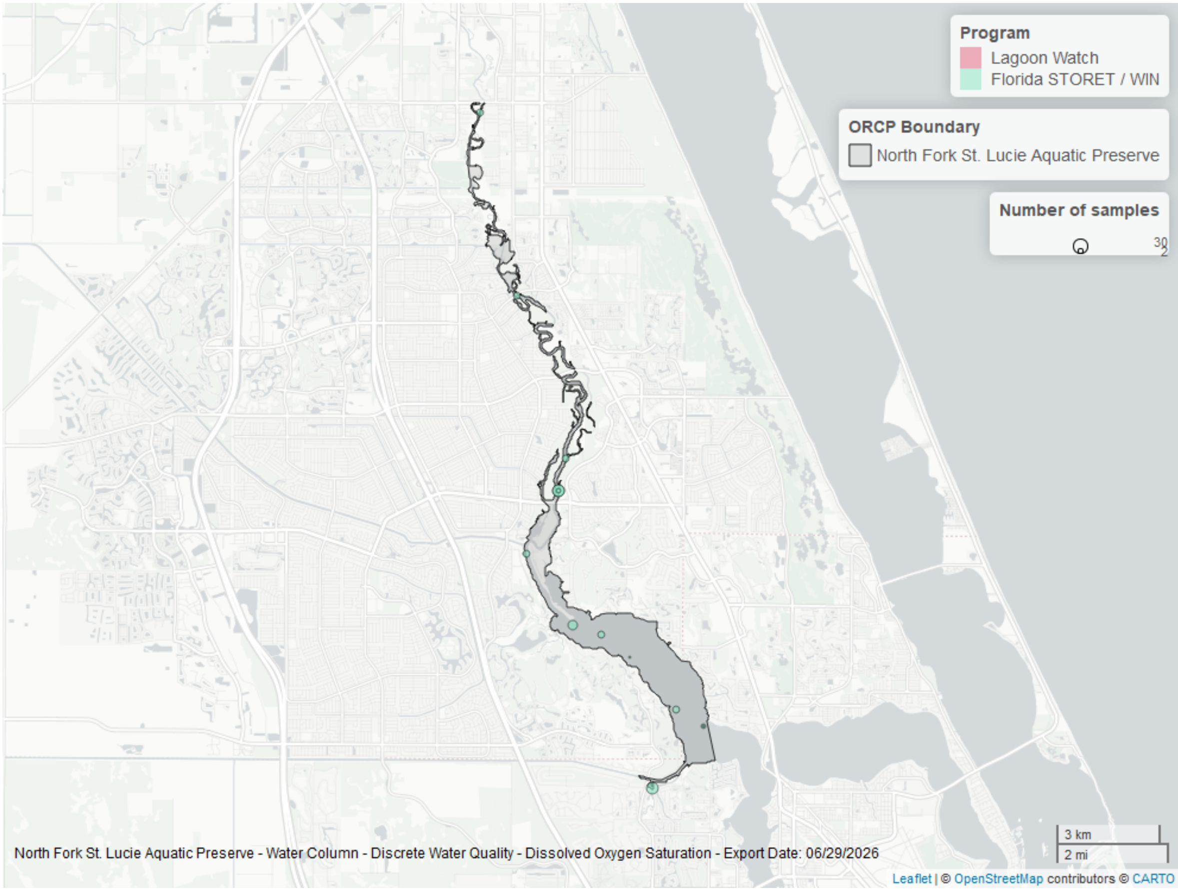

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

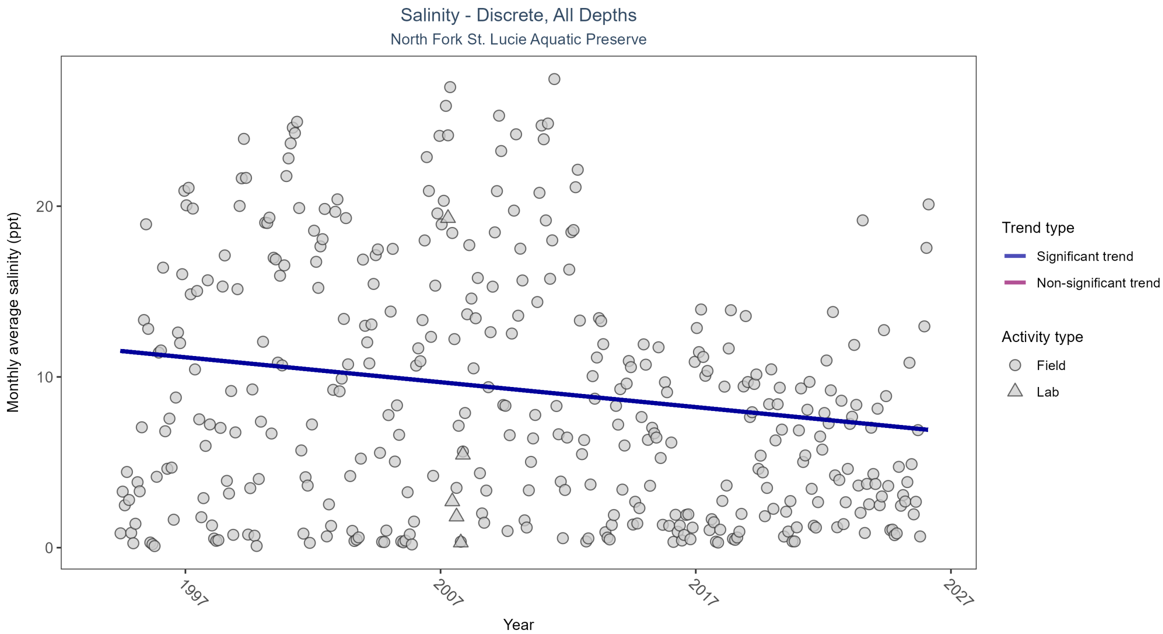

Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 5: Monthly average salinity decreased by 0.15 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Decreasing trend	7292	33	1994 - 2026	3.98	-0.17175	11.58691	-0.14585	0

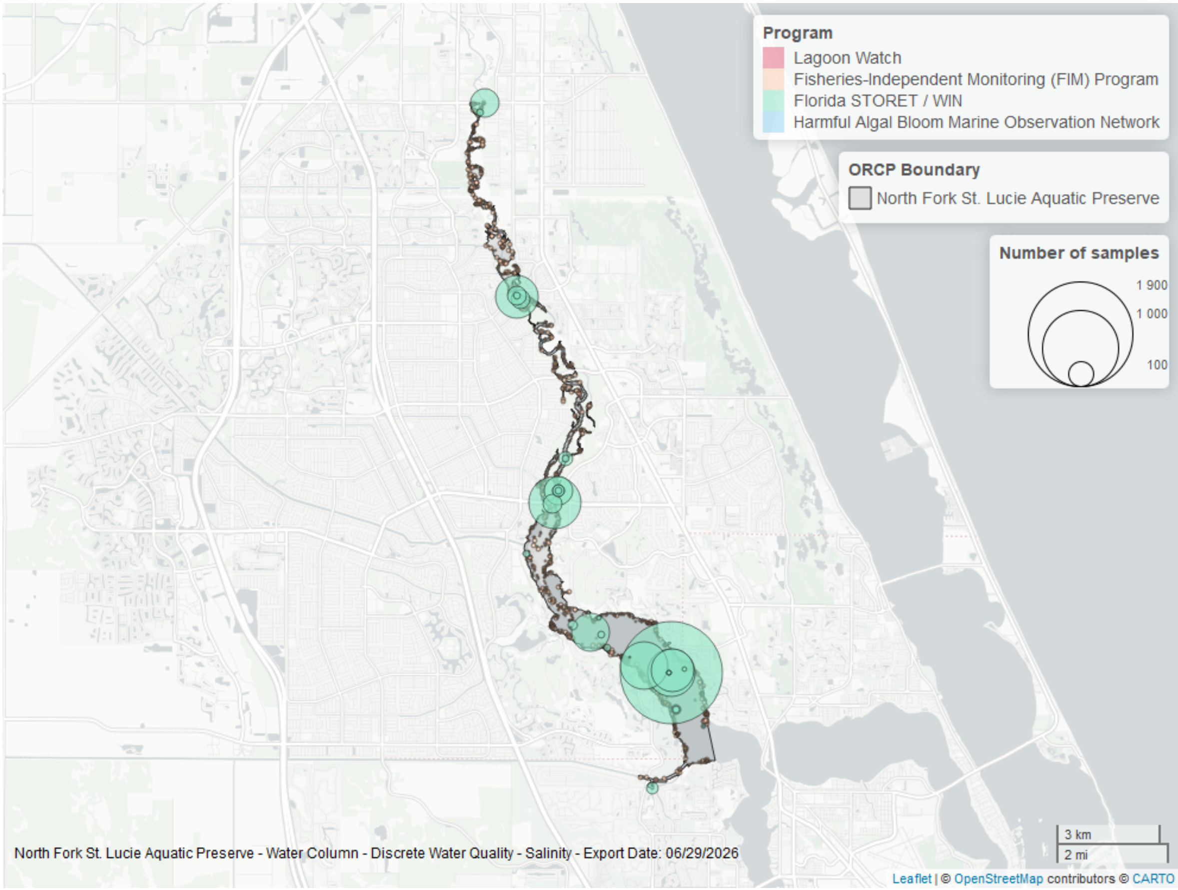

Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

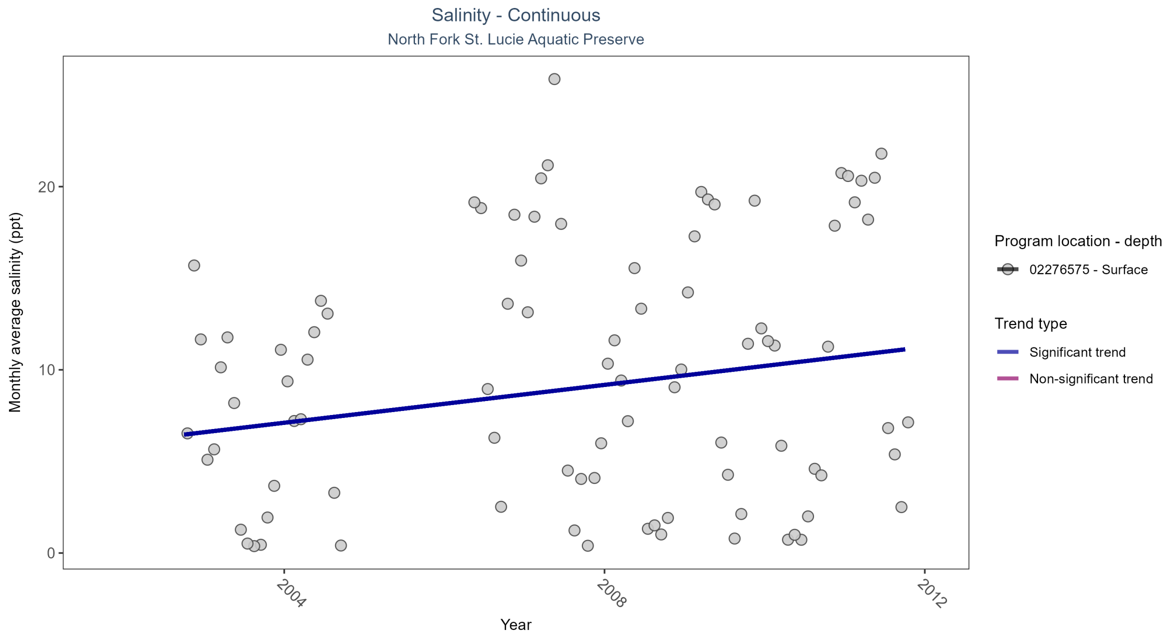

Figure 11: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 6: At one program location, monthly average salinity increased by 0.52 ppt per year.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02276575	Increasing trend	2666	9	2002 - 2011	9.1	0.22	6.08	0.52	0.02

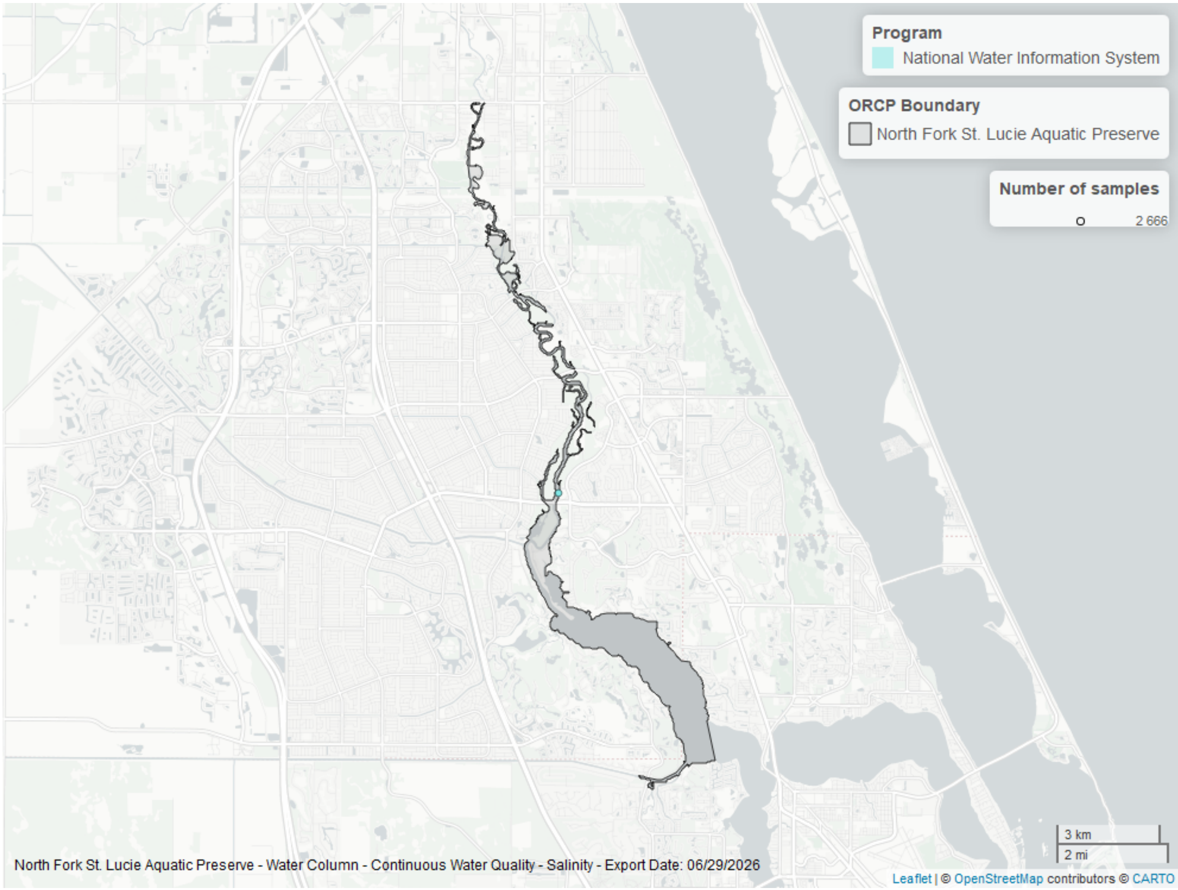

Figure 12: Map showing location of salinity continuous water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

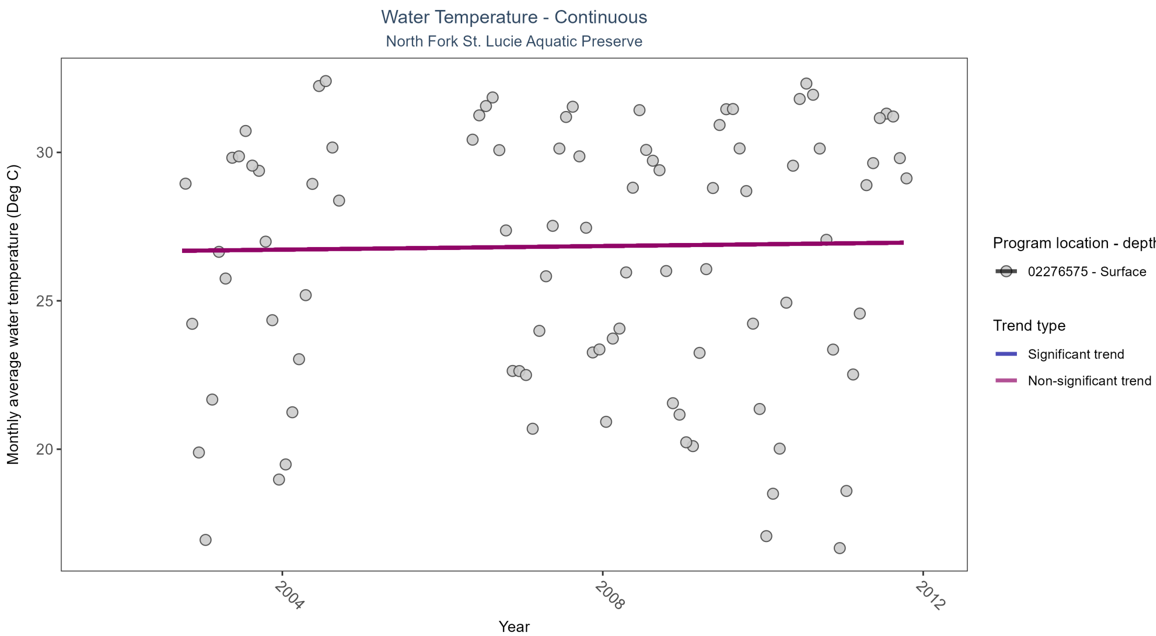

Figure 13: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: No detectable change in monthly average water temperature was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02276575	No detectable trend	2664	9	2002 - 2011	27.3	0.06	26.66	0.03	0.46

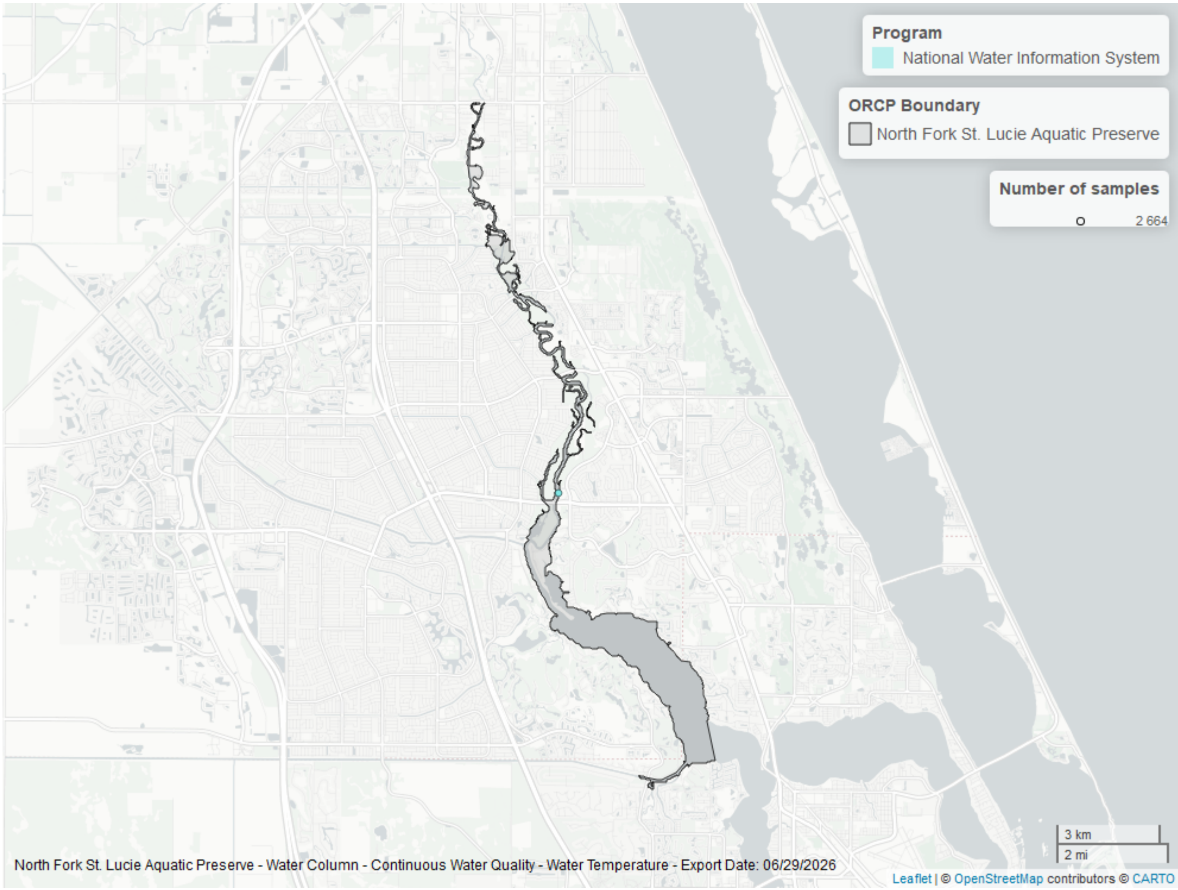

Figure 14: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

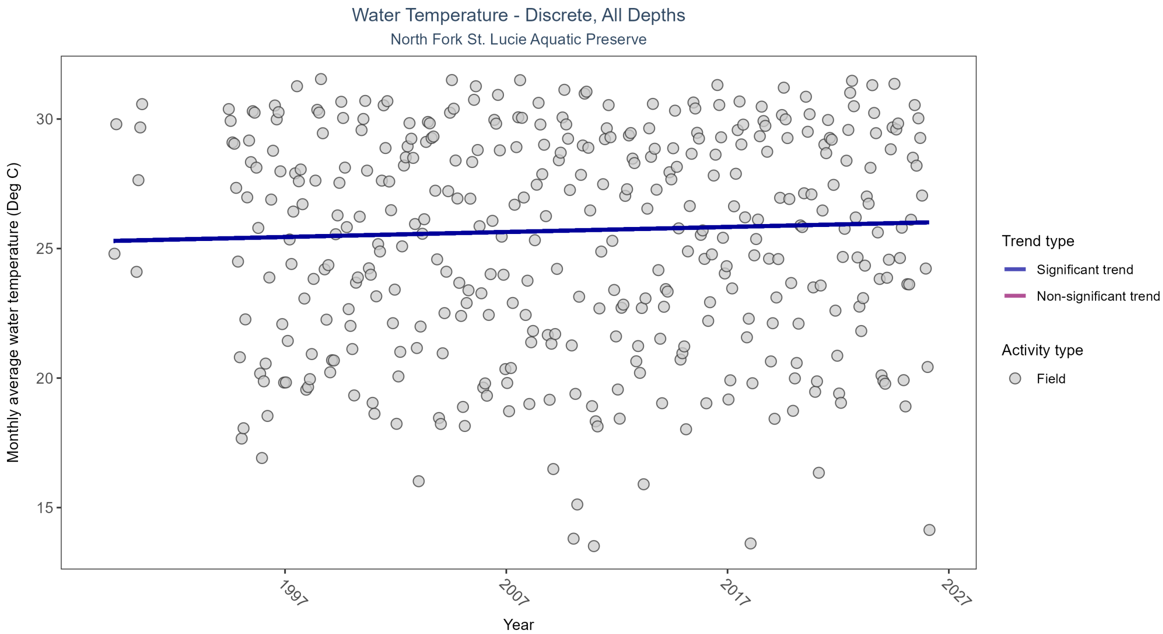

Figure 15: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 8: Monthly average water temperature increased by 0.02Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	7754	35	1989 - 2026	25.905	0.09502	25.2903	0.0194	0.0086

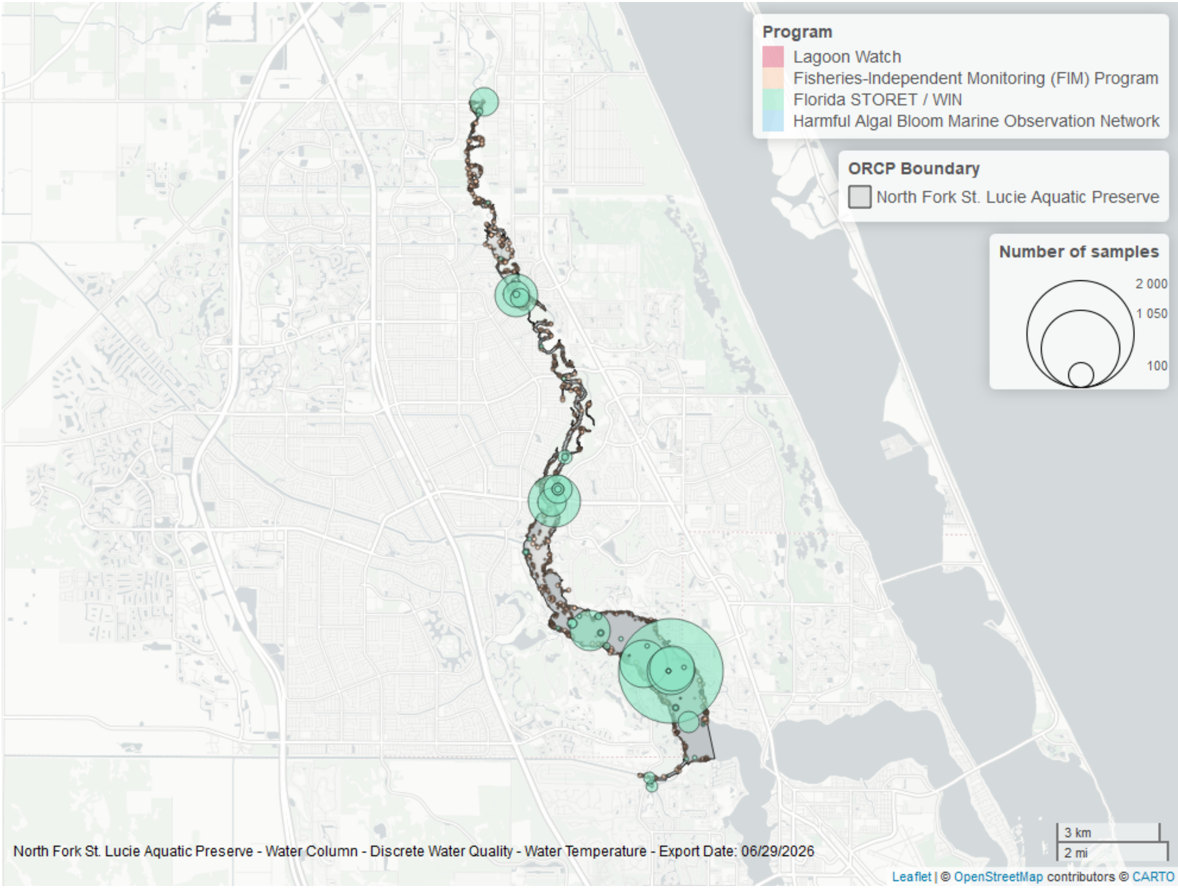

Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

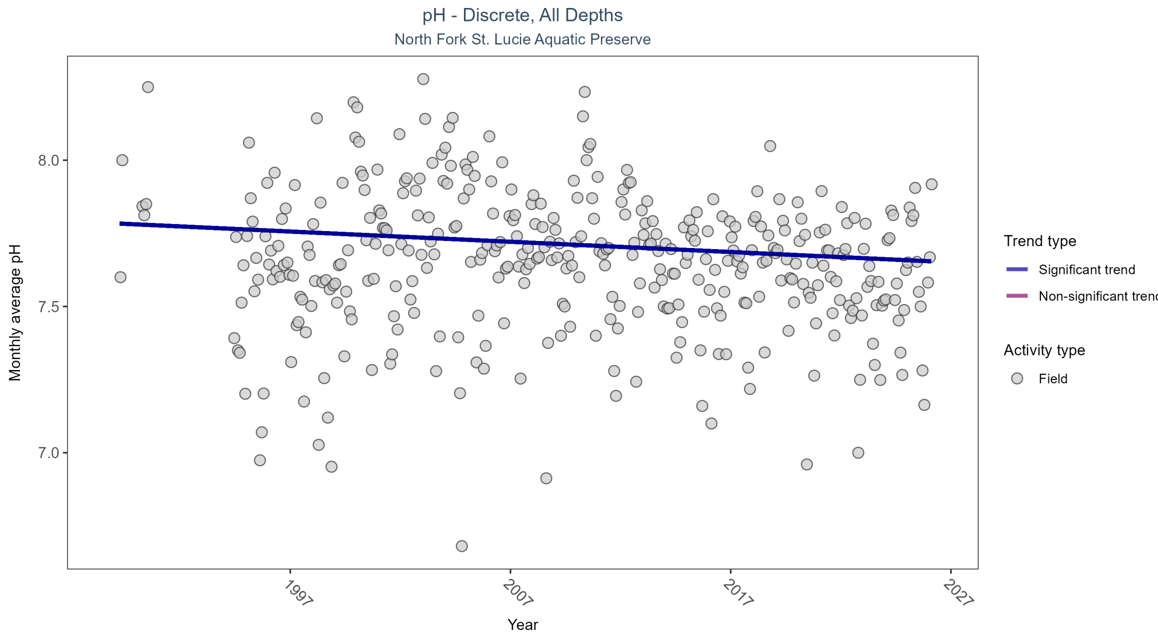

Figure 17: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 9: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	7659	35	1989 - 2026	7.62	-0.11859	7.78418	-0.0035	0.001

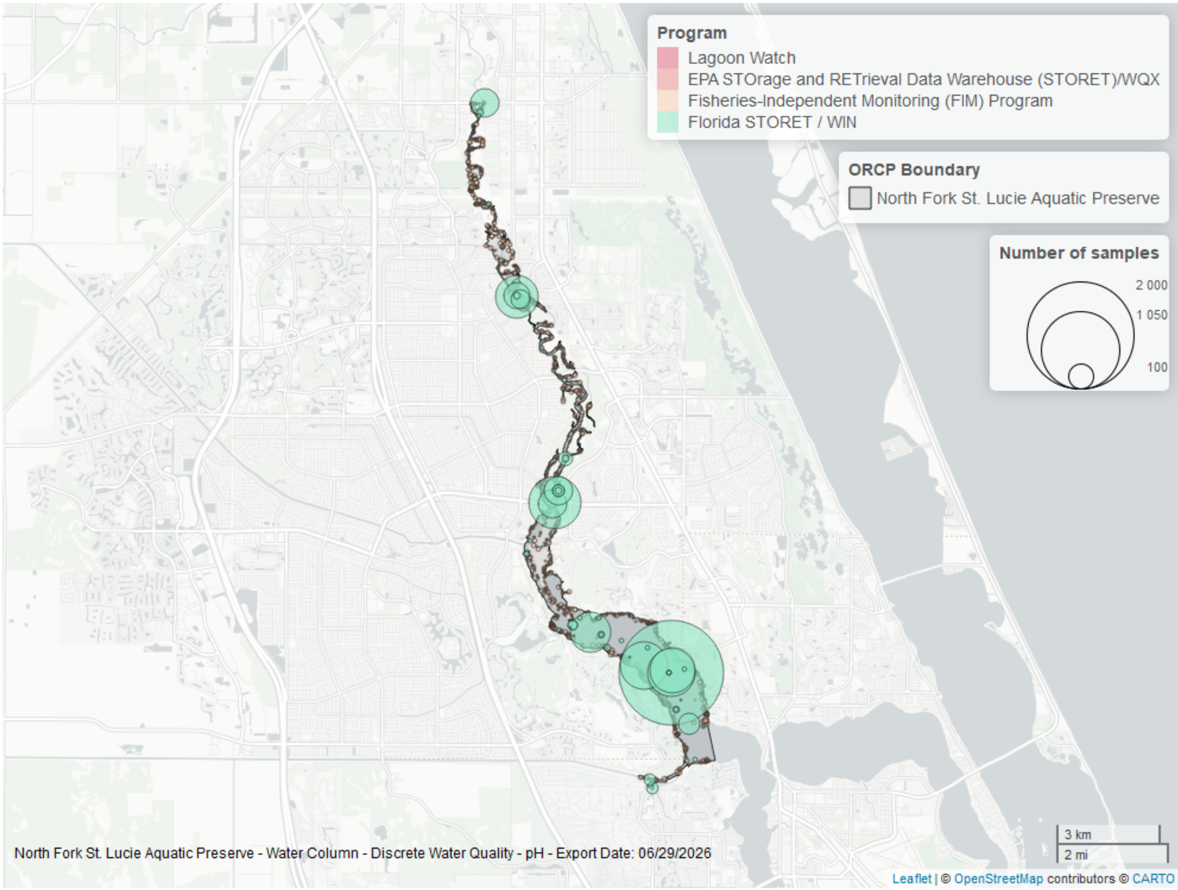

Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

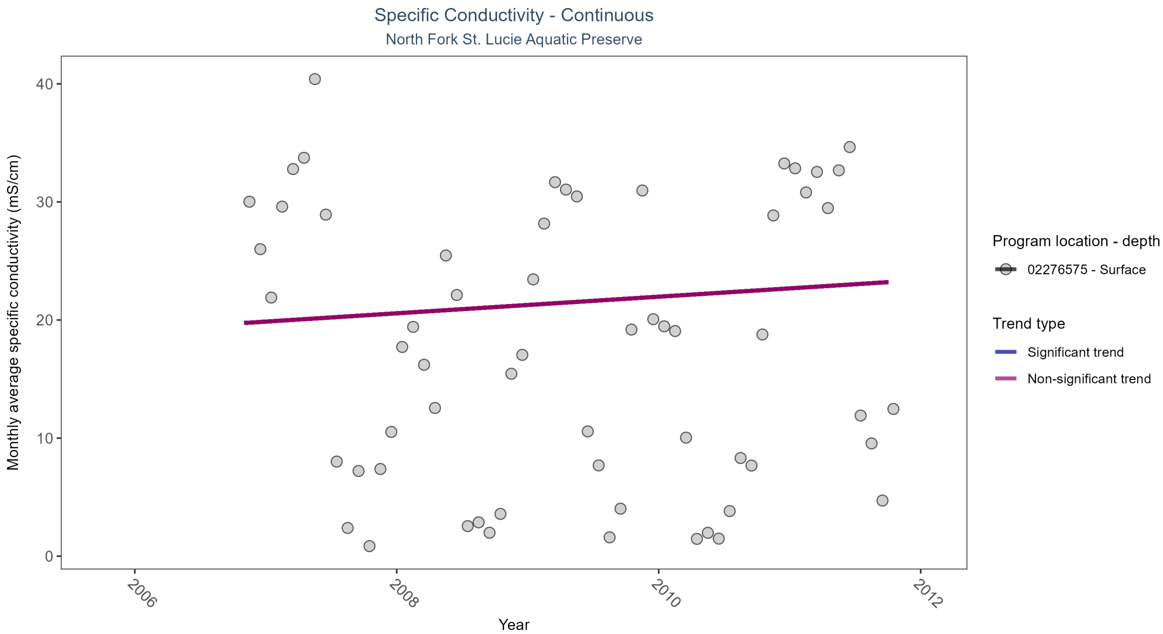

Figure 19: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: No detectable change in monthly average specific conductivity was observed at one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02276575	No detectable trend	1794	6	2006 - 2011	17.9	0.12	19.16	0.7	0.36

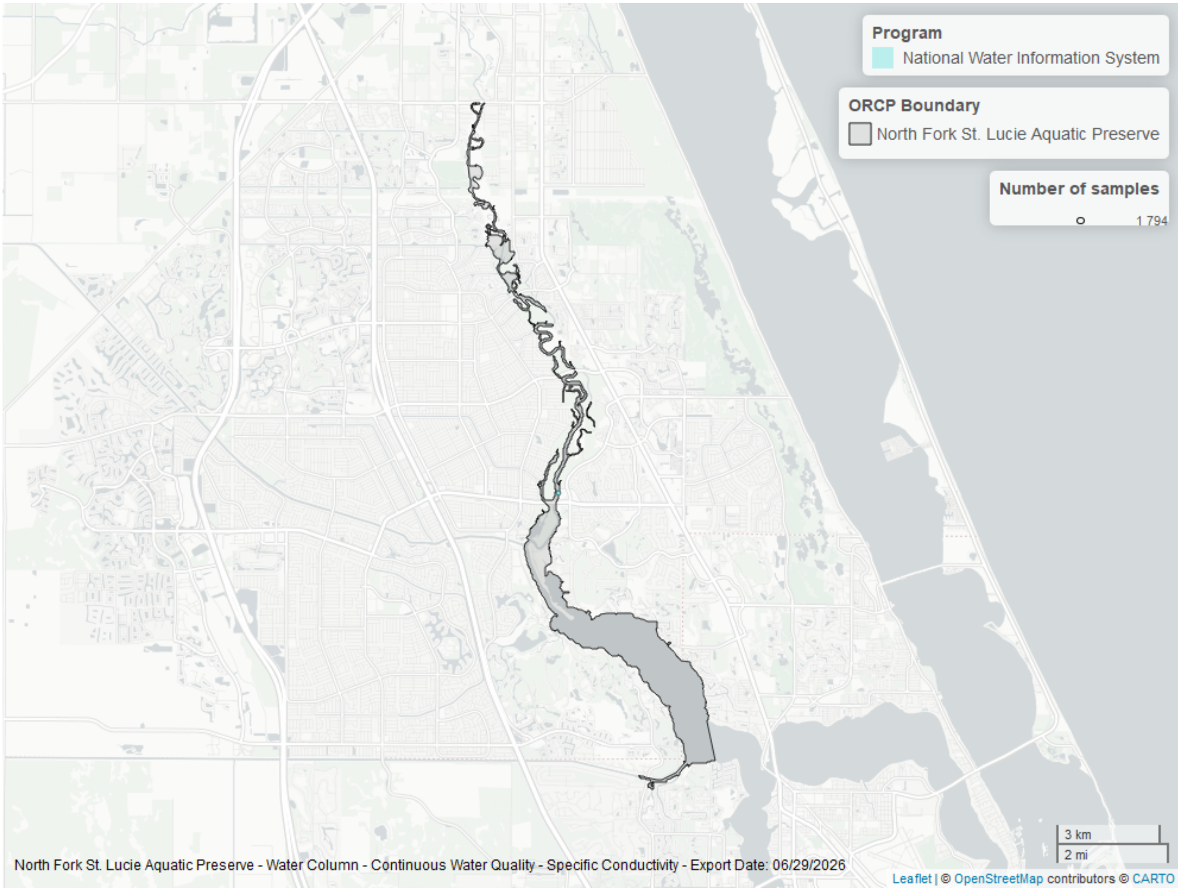

Figure 20: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Discrete

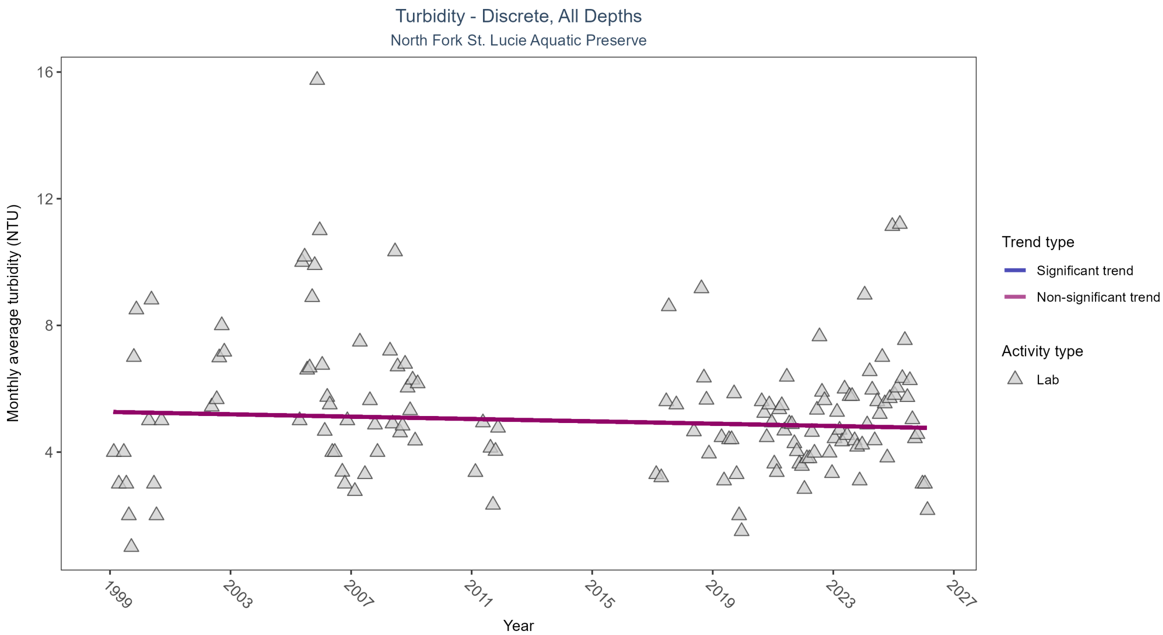

Figure 21: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 11: Turbidity showed no detectable trend between 1999 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	595	19	1999 - 2026	5	-0.02856	5.26728	-0.0185	0.5831

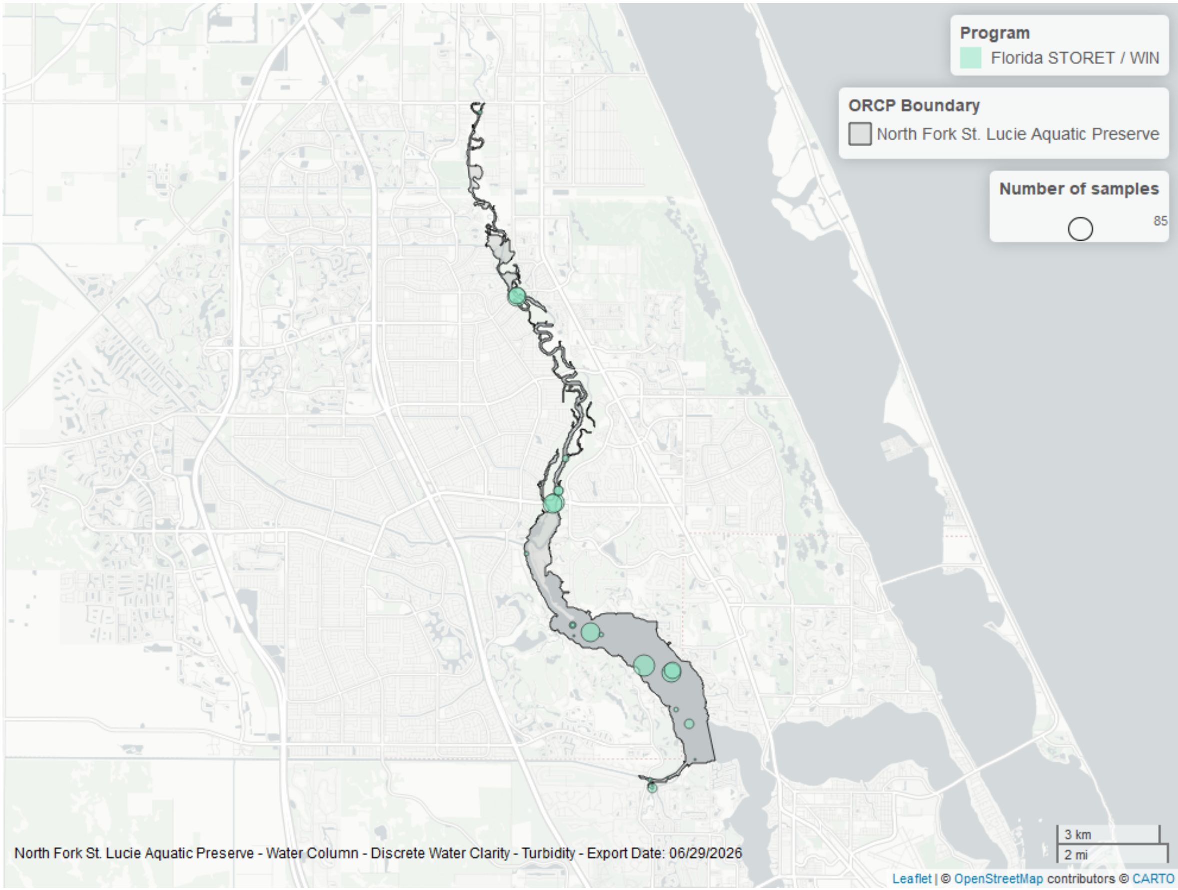

Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

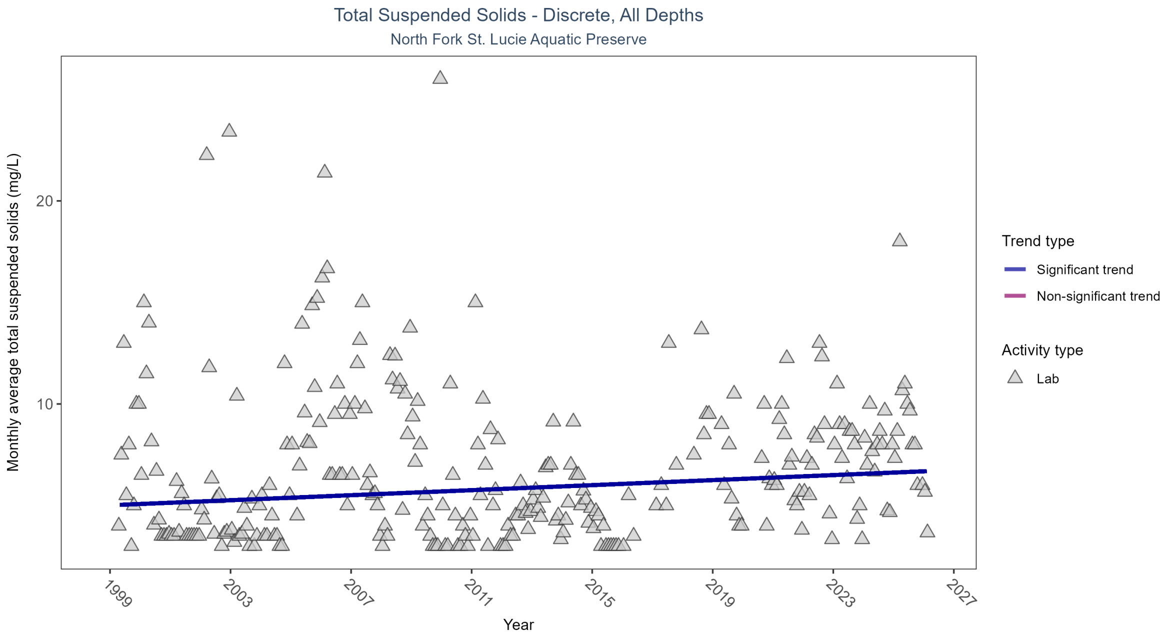

Figure 23: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 12: Monthly average total suspended solids increased by 0.06 mg/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1085	28	1999 - 2026	6	0.11998	5.01174	0.0619	0.0037

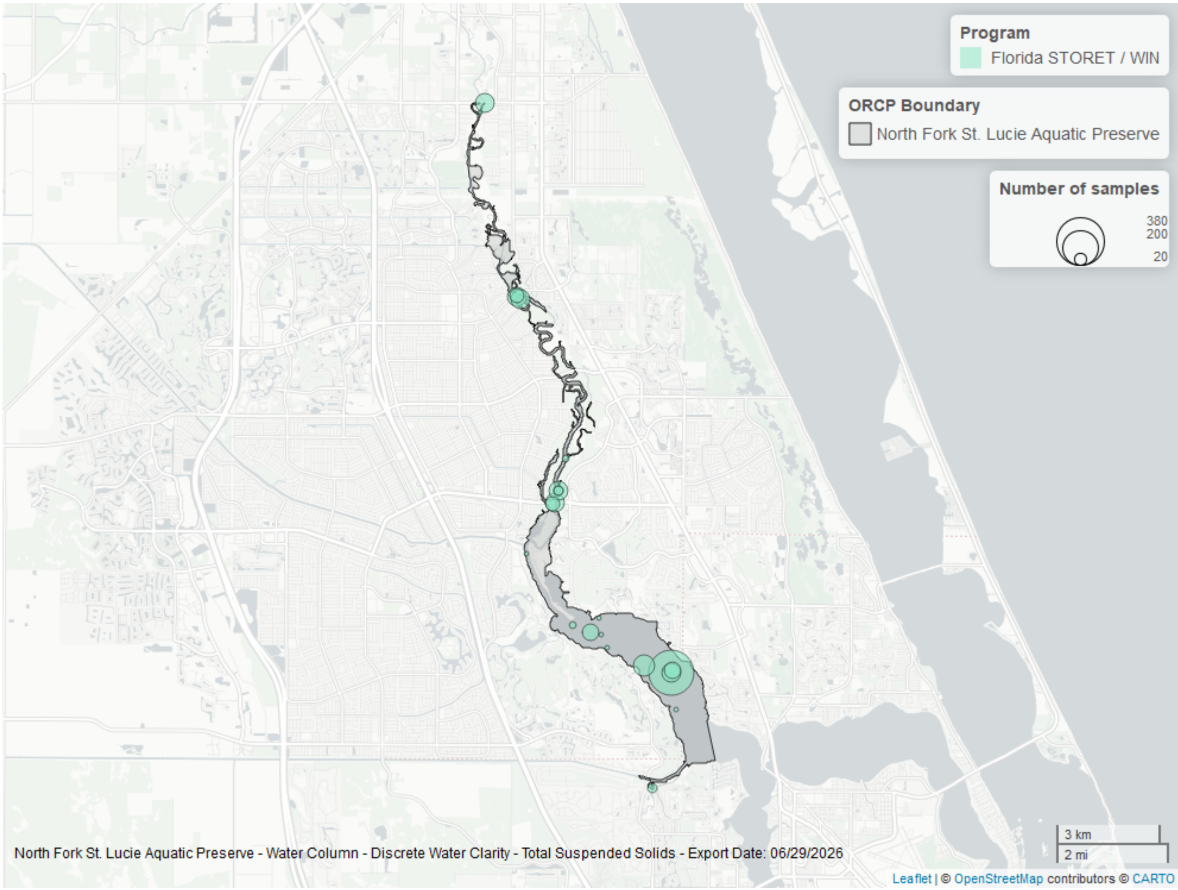

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

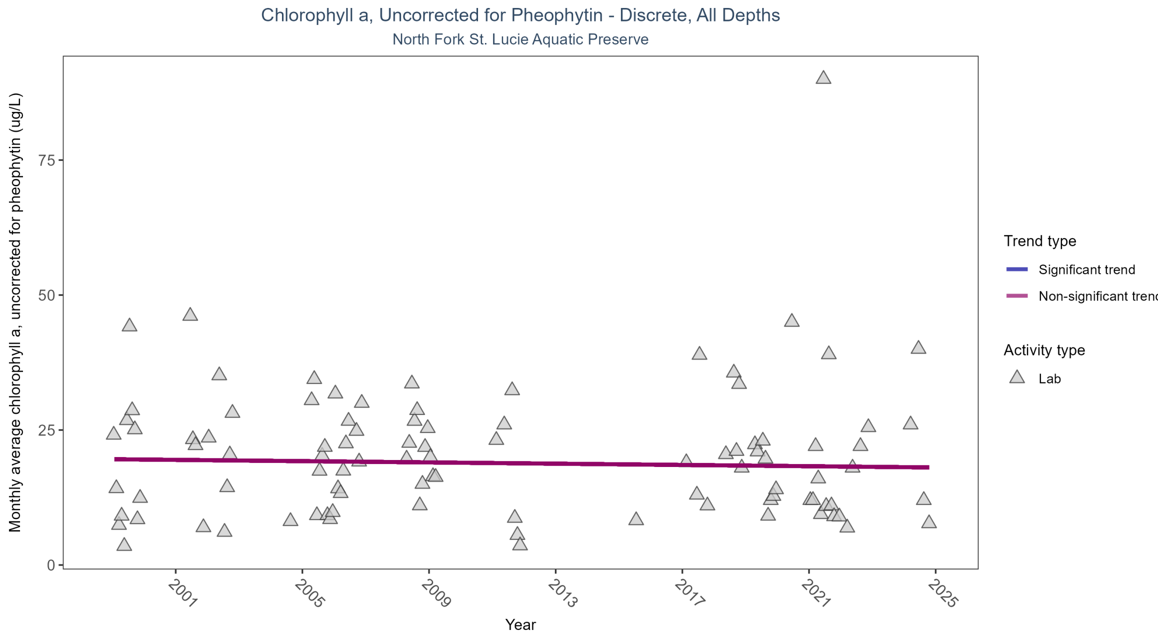

Figure 25: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 13: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 1999 and 2024.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	378	17	1999 - 2024	15.18	-0.07	19.59048	-0.05877	0.4435

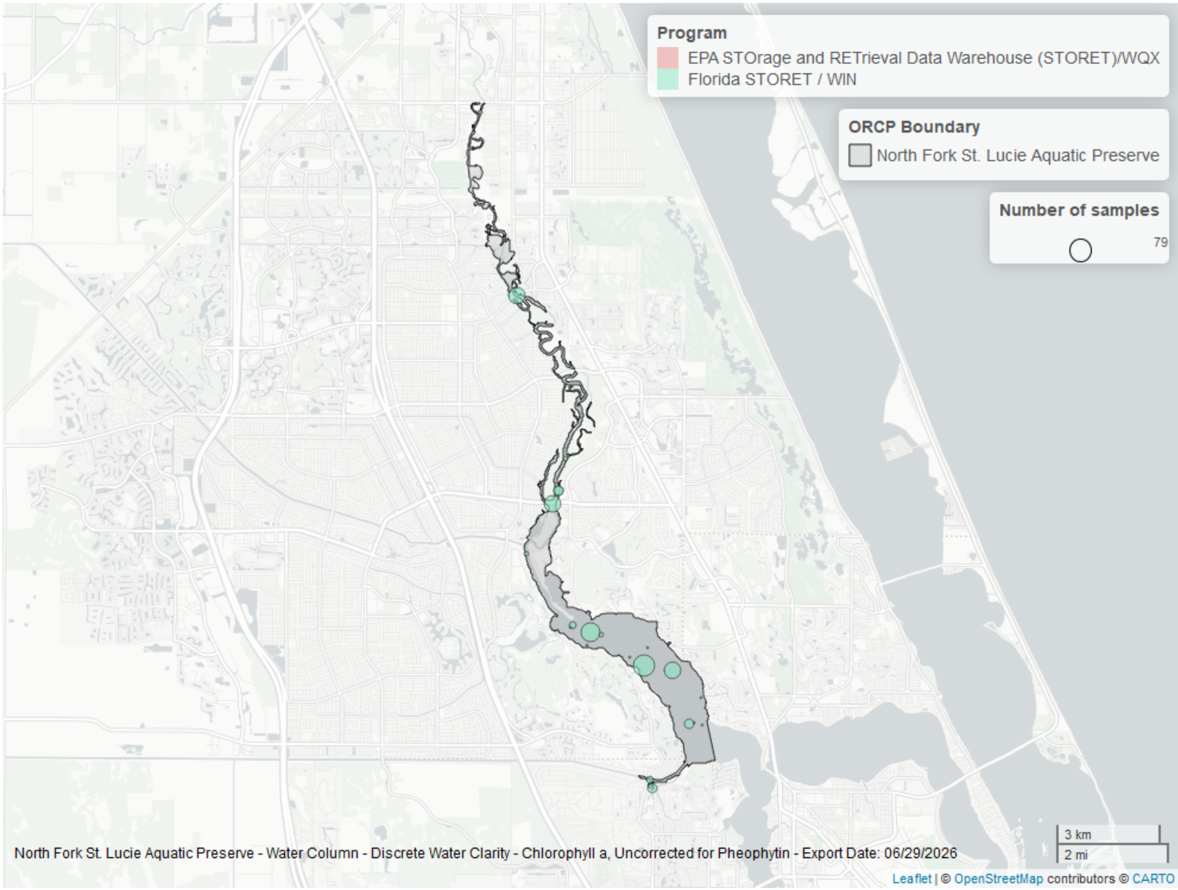

Figure 26: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

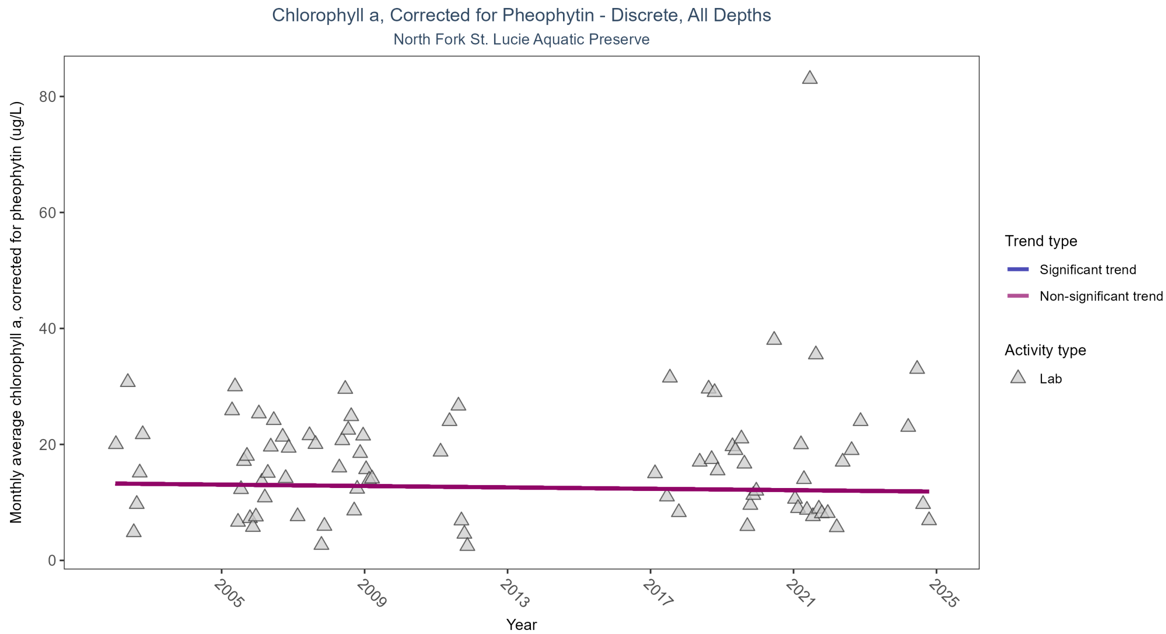

Figure 27: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 14: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2024.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	372	14	2002 - 2024	12	-0.03967	13.24892	-0.06	0.6464

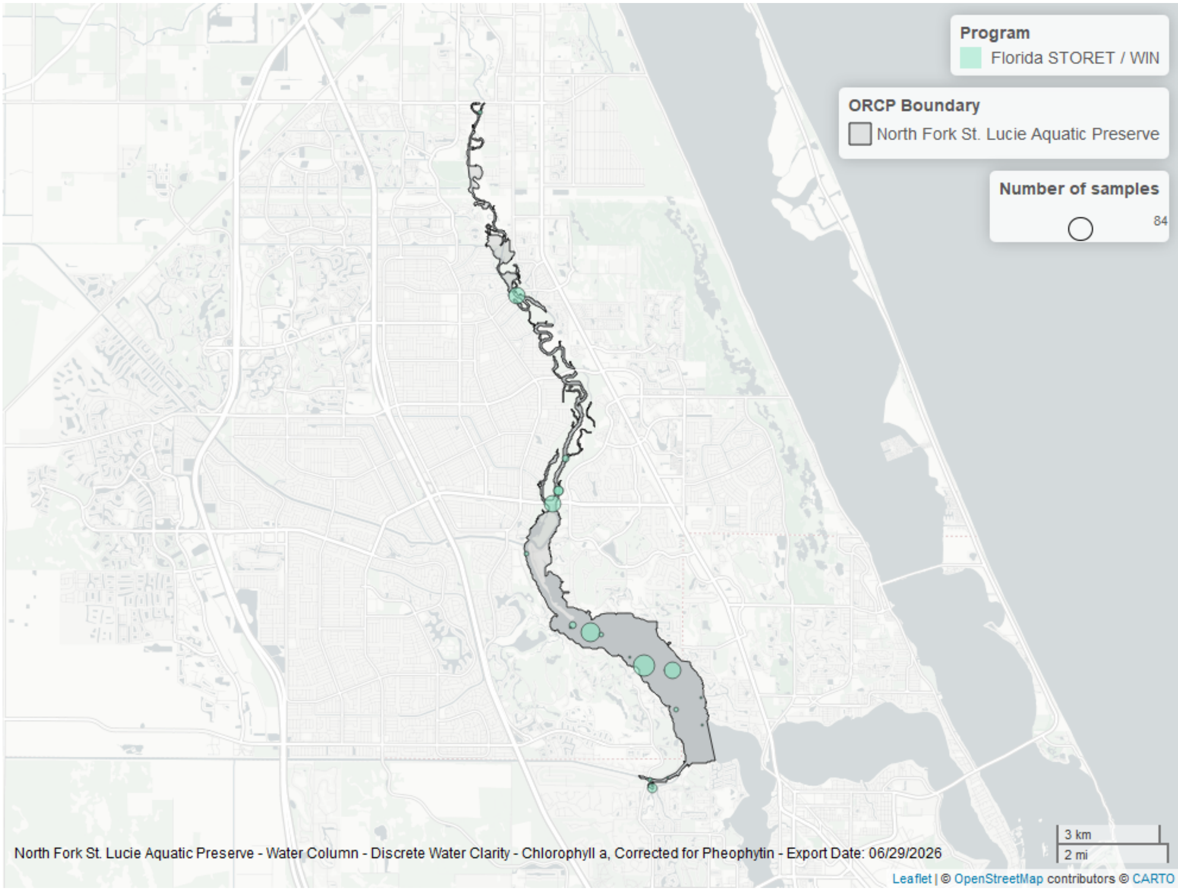

Figure 28: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

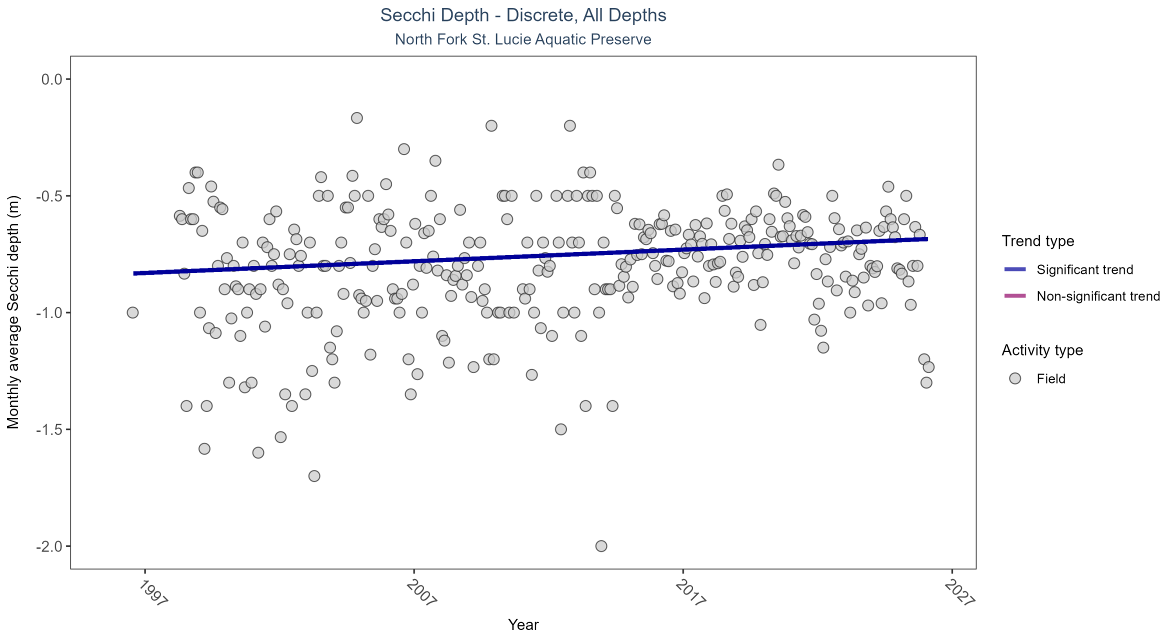

Figure 29: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 15: Monthly average Secchi depth became shallower by less than 0.01 m per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	2906	30	1996 - 2026	-0.7	0.12311	-0.83532	0.005	0.0016

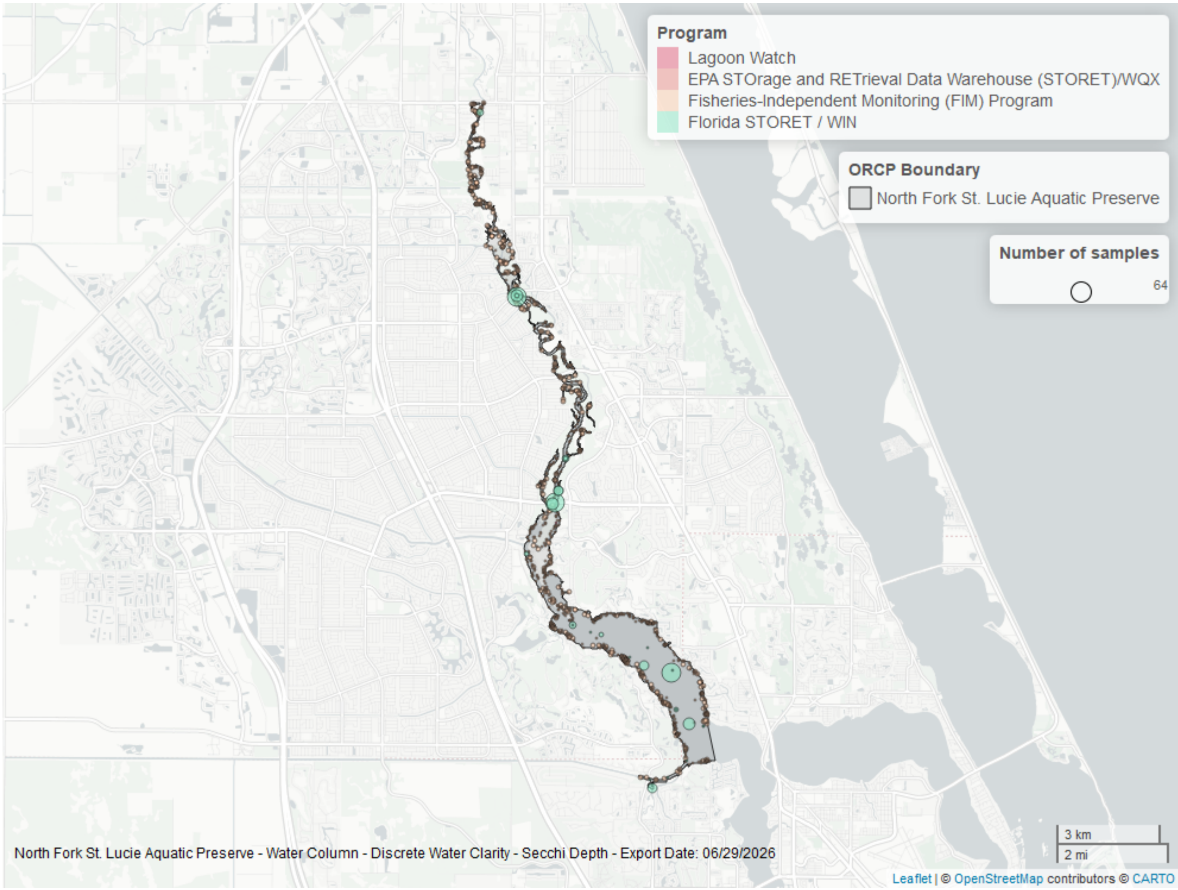

Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

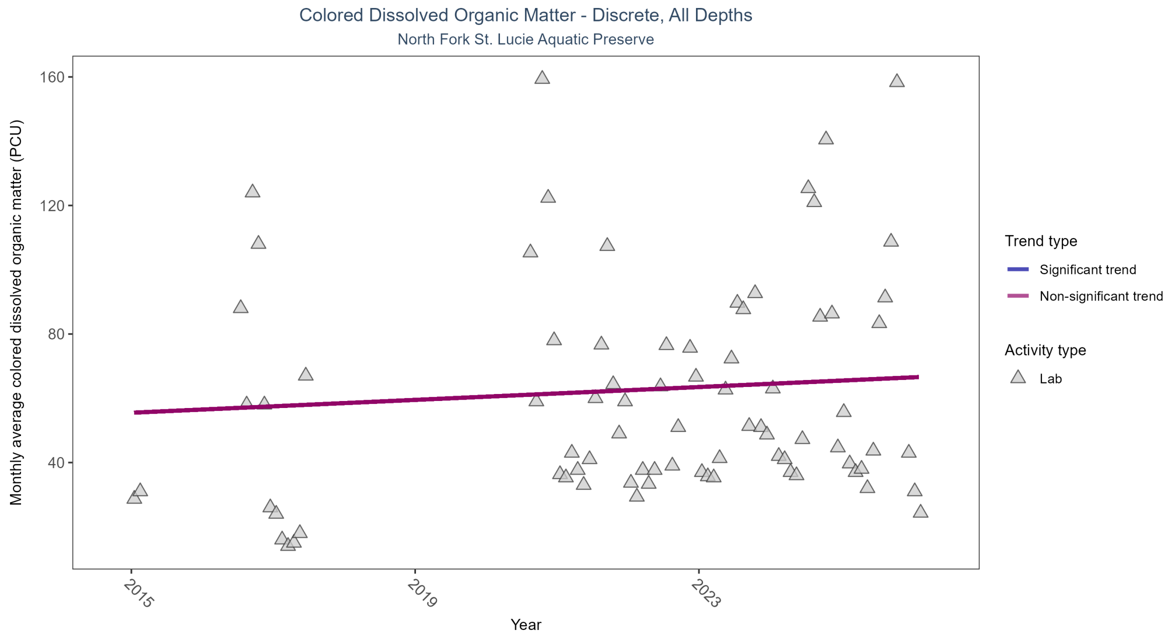

Figure 31: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 16: Colored dissolved organic matter showed no detectable trend between 2015 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	214	10	2015 - 2026	48	0.15419	55.5	1	0.1034

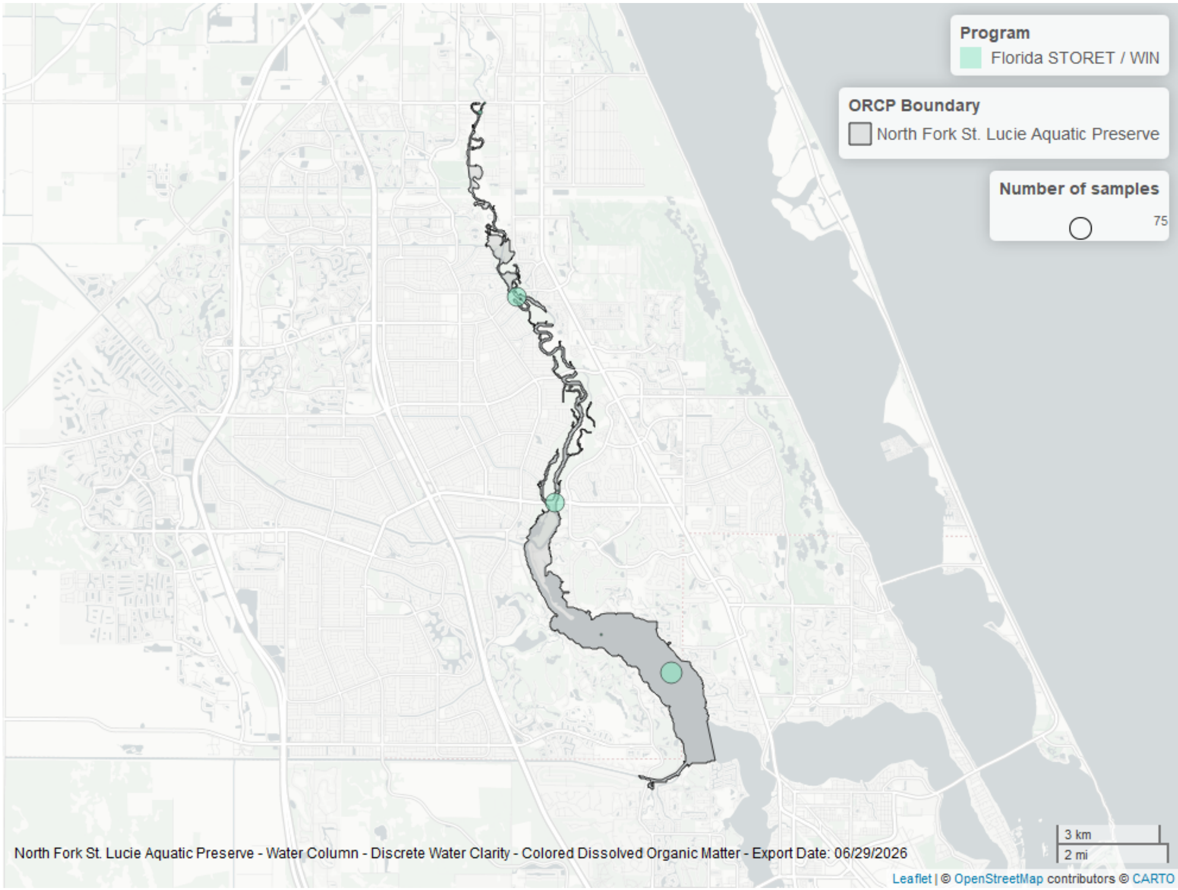

Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *North Fork St. Lucie Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.